

SECTION C

11 (**short**) In computer graphics, a curved surface may be approximated by a mesh of small triangles. Consider the following C++ data structures for representing a mesh comprising 11119 triangles sharing 5580 vertices.

```
class vertex {
public:
    double x; // x coordinate of vertex
    double y; // y coordinate of vertex
    double z; // z coordinate of vertex
};
class triangle {
public:
    int v1; // v_list index for vertex 1
    int v2; // v_list index for vertex 2
    int v3; // v_list index for vertex 3
};
vertex v_list[5580]; // vertex list
triangle t_list[11119]; // triangle list
```

(a) If a double occupies 8 bytes and an int occupies 4 bytes, how much memory is required to store the mesh? [3]

(b) Write a short C++ code segment to calculate the centroid of the first triangle in the mesh. [7]